PROFESSIONAL PROJECT REPORT

Fraud Alert Control Tower

Explainable fraud alert prioritization, investigator queue control, threshold economics, and governance evidence.

Prepared by	Mirza Saif Baig
Date	June 21, 2026
Project	JPMorgan-inspired explainable fraud alert prioritization
Status	Submission-ready prototype using synthetic data

1. Executive Summary

This project was built to solve an explainable fraud alert prioritization problem. I interpreted the brief as an operational analytics challenge: a fraud team cannot review every transaction, so the system must rank the riskiest cases, explain why each case is risky, and show how the threshold affects investigator workload.

The final deliverable is an end-to-end fraud analytics workflow covering data cleaning, feature engineering, model comparison, threshold selection, explainability, governance, and a deployed Vercel-style analyst app.

Final conclusion

The project is not only a fraud classifier. It is a decision-support control tower that connects fraud scoring to review queues, investigator explanations, false positives, and governance controls.

2. Project Brief And My Interpretation

The assigned problem was to build an explainable fraud alert prioritization system. My first decision was to avoid treating it as a simple fraud vs legitimate prediction task. In a bank, the bigger question is which alerts should be reviewed first when investigator capacity is limited.

- Score transaction-level fraud risk.
- Prioritize a review queue instead of only producing a model score.
- Explain the main drivers behind each high-risk case.
- Tune thresholds around investigator capacity and false-positive load.
- Document limitations and responsible-use controls before production use.

3. Project Journey

Phase	What I did	Why it mattered
1. Understand the brief	Reframed the task as fraud operations and review-capacity prioritization.	This made the work more realistic for a banking fraud team.
2. Prepare the	Joined transaction, customer, and	A model is only useful if the data

data	merchant information; fixed known data issues; prevented leakage.	pipeline is trustworthy.
3. Build features	Selected the top 10 explainable features using training-only mutual information.	Kept the model defensible and easy to connect to the app.
4. Compare models	Tested weighted models and evaluated them using fraud-focused metrics.	Accuracy is misleading when fraud is rare.
5. Deploy workflow	Built a Vercel-style control tower with scoring, queue, metrics, governance, and export.	Turned the analysis into something a reviewer can operate and assess.

4. Data Cleaning And Preparation

The modelling table was created at transaction grain, meaning one row represented one transaction. Customer and merchant fields were joined in as supporting context. The cleaning stage focused on keeping records, flagging data quality issues, and removing fields that could leak the answer.

- Parsed transaction timestamps into hour, day-of-week, month, and weekend features.
- Normalized country values, including mapping GB to UK for consistency.
- Left joined customer and merchant profiles so transaction records were preserved.
- Created missing-profile flags rather than dropping unmatched customer records.
- Flagged broken tenure values instead of treating them as valid numeric tenure.
- Excluded IDs, target fields, and the existing alert flag from model inputs.

5. Feature Engineering And Selection

Feature selection was performed using training-only mutual information. This avoided selecting features based on validation or test data and kept the final app aligned with the same input fields used by the modelling workflow.

Rank	Feature	Reason it is useful
1	geo_distance_km	Captures unusual transaction location distance.
2	txn_country	Captures country-level transaction context.
3	synthetic_identity_score	Flags possible identity-profile risk.
4	merchant_risk_score	Adds merchant-level risk signal.
5	channel	Separates P2P, wire, card, and online behavior.
6	txn_hour	Captures time-of-day transaction pattern.
7	device_risk_score	Adds device-level fraud risk context.
8	merchant_profile_risk_score	Adds broader merchant profile signal.
9	transaction_amount_usd	Captures transaction value risk.
10	amount_log1p	Reduces skew from transaction amount.

6. Modelling And Threshold Strategy

The modelling stage compared several weighted models because the fraud class was rare. I did not treat accuracy as the main evidence. Instead, I focused on ranking quality, precision, recall, false positives, and review count.

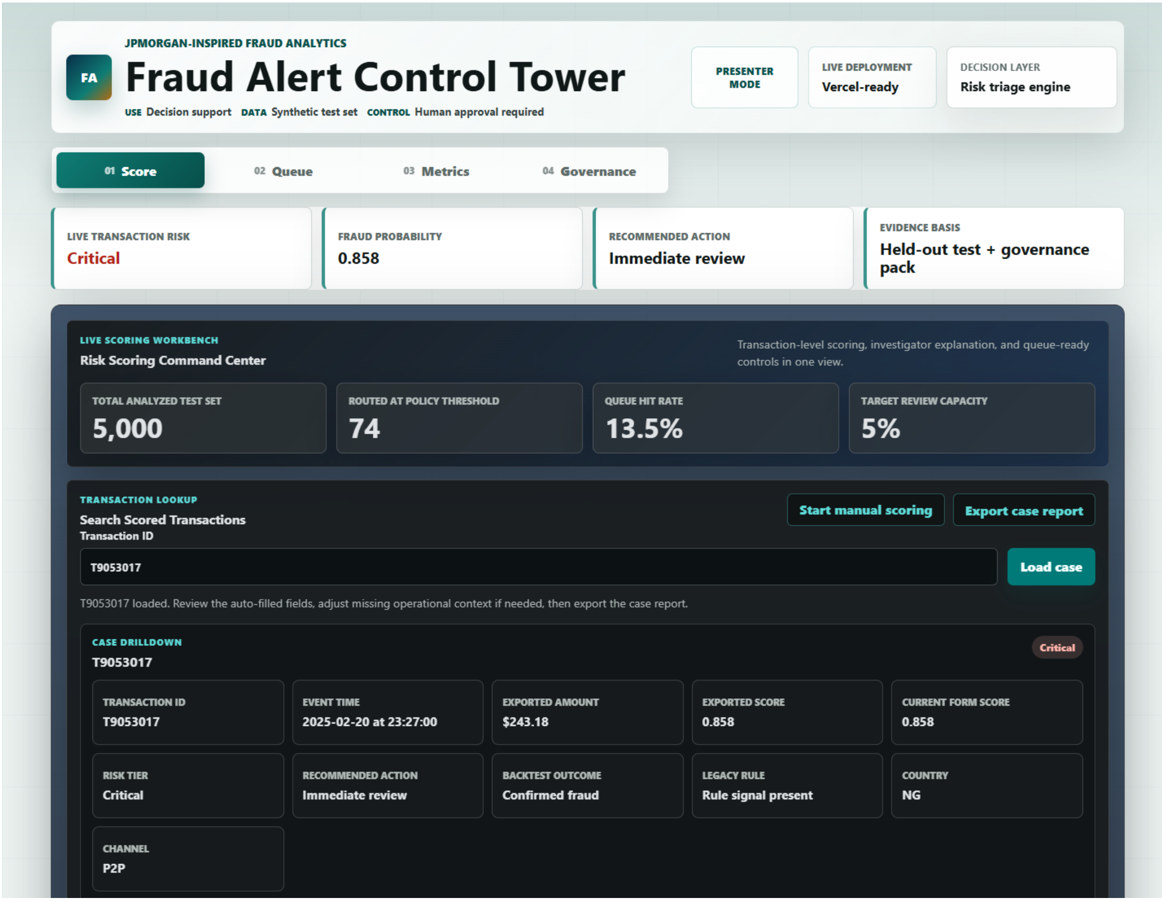
Metric	Selected operating result	Interpretation
Recommended threshold	0.798	Prototype threshold selected around 5% review capacity.
Review count	74 of 1,000 test transactions	Actual test queue rate was 7.4%.
Queue hit rate	13.5%	Share of reviewed cases that were confirmed fraud.
Fraud capture	25.0%	Share of all frauds captured in the review queue.
True positives	10	Fraud cases caught at the selected operating point.
False positives	64	Legitimate cases requiring review before action.

The key conclusion is that the threshold is a business-control decision. Lower thresholds increase fraud capture but create more false positives. Higher thresholds reduce workload but risk missing fraud.

Tier	Probability band	Analyst meaning
Critical	p >= max(0.820, review threshold)	Emergency review before any customer-impacting decision.
High	review threshold <= p < Critical band	Manual investigator review.
Medium	0.350 <= p < review threshold	Monitor closely and escalate if behavior repeats or another signal appears.
Low	p < 0.350	Likely legitimate; keep score and reasons in the audit trail.

7. Deployed Analyst App

The final app was built as a Vercel-style static application so reviewers can open it in a browser without running a Python server. The app has four main sections: Score, Queue, Metrics, and Governance.



Score view: transaction lookup, manual scoring, risk decision, explanation, and case export.

App section	Purpose
Score	Search or manually enter a transaction and receive risk probability, alert tier, action, explanation, and CSV export.
Queue	Show the investigator review queue, top cases, rule-gap cases, triage lanes, and control checks.
Metrics	Translate model performance into fraud capture, false positives, review cost, and threshold trade-offs.
Governance	Summarize model card, limitations, approval gates, monitoring controls, and responsible-use policy.

8. Explainability And Investigator Use

The app explains the score by comparing each transaction with typical reference values. If a transaction has unusually high geographic distance or amount, the app shows how that signal increased model risk. If a feature lowers risk, the app presents it as a protective signal.

Important wording control: the explanation describes model behavior, not guaranteed causality. It means the model treated a feature value as risk-increasing or risk-reducing; it does not prove that the feature caused fraud.

- High-risk cases receive investigator actions such as validation, step-up verification, or manual review.

- Low-risk cases can be monitored with audit trail only unless behavior changes.
- The exportable CSV case report gives transaction facts, risk score, explanation, next steps, and governance notes.

9. Governance And Responsible Use

Because this is a banking-style fraud project, governance was treated as part of the deliverable rather than an afterthought. The model is positioned as decision support only. It should not automatically freeze accounts or create customer-impacting decisions without human review.

Control	How it was handled
Leakage prevention	Removed IDs, target fields, and the old alert flag from model inputs.
Human review	App recommends review actions but does not automate blocking.
Threshold governance	Threshold selected with investigator capacity and false-positive load in mind.
Monitoring	Defined queue size, hit rate, fraud capture, false positives, drift, and segment review.
Limitations	Synthetic data means production use would require real-data validation and approval.

10. Enterprise Readiness Roadmap

I compared the prototype with what a bank-grade fraud platform would require and separated the roadmap into three levels. This makes the limitation clear without weakening the project: the current version demonstrates the analyst workflow, while the next stages describe what would be needed for a controlled pilot and production approval.

Prototype now: transaction scoring on synthetic data, risk tiers, investigator queue, local reason codes, threshold economics, CSV case export, and governance evidence.

Controlled pilot: probability calibration against observed fraud rates, real SHAP explanations, segment validation by country/channel/amount band, investigator feedback capture, and weekly drift monitoring.

Production requirements: live transaction streams, customer and account history, device and login behavior, graph/network fraud signals, role-based access, audit logging, security review, and formal model-risk approval.

11. Changes Made After Feedback

After receiving feedback, I improved the project to make it easier for a tutor or JPMorgan-style reviewer to understand and evaluate. The main changes focused on clarity, professional presentation, and reviewer usability.

Feedback theme	Change made
PPT needed a stronger story	Created a new journey-based deck explaining how the project started, evolved, and reached its conclusion.
App needed clearer analyst flow	Improved opening state, labels, sections, and risk analyst wording.
Export report needed to be more useful	Changed the case report export to CSV with transaction facts, decision evidence, and governance notes.

Tutor may ask how it was built	Added optional Presenter Mode so build notes appear only when needed.
Submission folder needed focus	Kept final submission materials project-focused and separated from learning notes.

12. Project Report Card

Review area	Evidence in this project
Fraud relevance and feature engineering	Top-10 fraud features, transaction/customer/merchant joins, leakage prevention, and explainable inputs.
Model evaluation and threshold discipline	Weighted model comparison, held-out test metrics, threshold capacity ladder, false-positive analysis.
Explainability and governance	Local explanations, model card, threshold memo, responsible-use summary, monitoring controls.
Operational dashboard usefulness	Score, Queue, Metrics, Governance, transaction lookup, case drilldown, and CSV export.
Business communication	Alert economics, review cost framing, fraud capture trade-off, final PPT, and professional report.

13. Final Recommendation

For the final review, I would present the project as an end-to-end fraud analytics control tower. The strongest message is that the work starts with data cleaning and modelling discipline, but the final value is operational: helping investigators decide what to review first, why it is risky, and how the review threshold affects cost and workload.

Two-minute closing pitch

I built an explainable fraud alert prioritization system that cleans and joins transaction, customer, and merchant data; selects top risk features; compares fraud models; tunes the threshold around investigator capacity; and deploys a Vercel-style control tower with scoring, queue, metrics, governance, and case export.